CTRL FREQ

Losing Ctrl since 666!

μ-LSe - Assembly Instructions

Part 1 — Hardware

Intro

Congratulations on purchasing your new life companion! Below you will find instructions on how to assemble and bring your new pet to life.

We are truly delighted that you chose us - but let's face it, you had no choice.
Here at **CtrlFreq** we share and cherish the universal values of late-stage capitalism, including but not limited to: endless exploitation of natural and human resources, tax evasion, putting profit before people, and brainwashing (aka advertising).

Tools & Skills

If this is your first time soldering, it's strongly recommended to follow a few tutorials, such as this one.

Required tools: soldering station, wire cutter, cross screwdriver

Helpful (but optional) tools: wire stripper, hot glue gun, third hand, multimeter

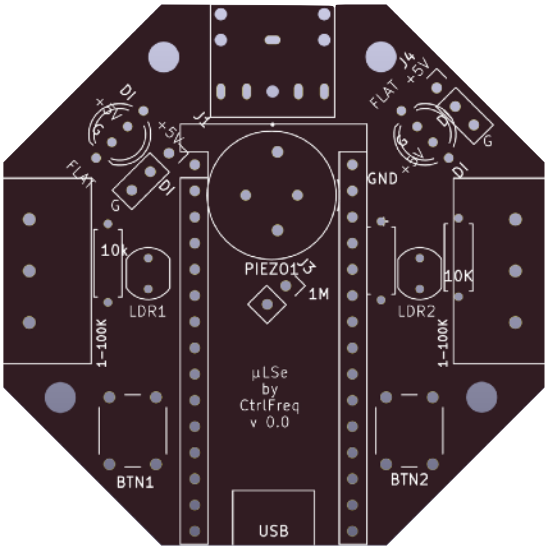
When soldering, always work in a well-lit and ventilated room. We can't be held responsible for anything. We have good lawyers, and we will win any court case. Don't even think about it.

Getting Started

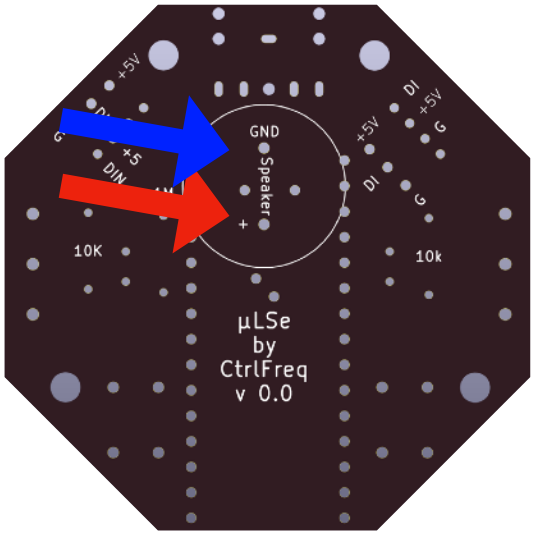
Soldering is like building a house - you start with the foundations and work your way up. Expensive, time-consuming, and not for everyone.

Start by identifying the **top** and **bottom** sides of the skeleton (aka PCB).

TOP



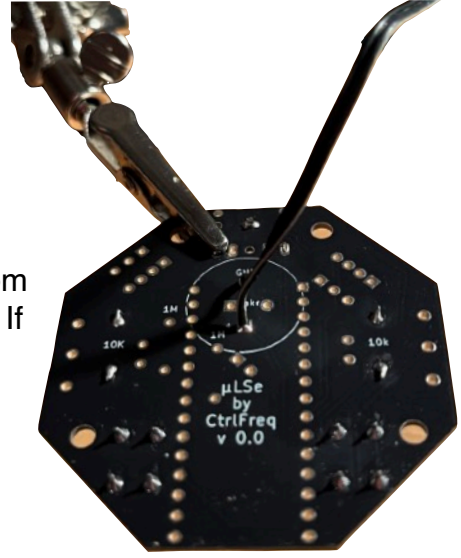
BOTTOM



Step 1 — Resistors

Solder the two **10 kΩ resistors** (marked as *10k*) on the top side of the PCB. Polarity doesn't matter - they can go either way. After soldering, cut the resistor legs with wire cutters so they don't get in the way.

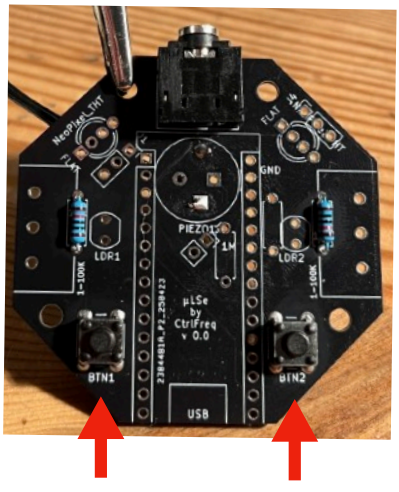
Step 2 — Speaker Wire



Solder a **two-core wire** (about 5 cm long) on the bottom of the PCB at the points marked **Speaker GND** and **+**. If possible, use a **black wire for GND** and one with a **white stripe for +**. We'll use the other ends of these wires later to connect the mini speaker (or piezo) output.

Adding a **piezo** as an extra touch sensor input is optional. If you decide to add it, solder the **1 MΩ resistor**; otherwise, you can skip this step.

Step 3 — Buttons

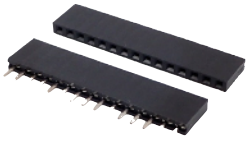


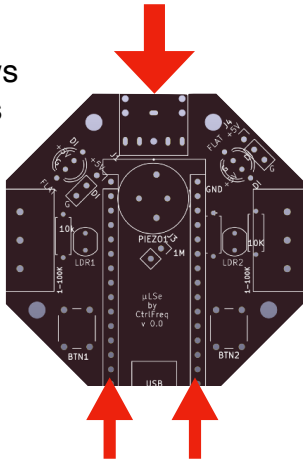
Back on the top side, solder the two **mini buttons**.

A button is a digital input - it allows our creature to sense whether something *is* or *isn't*. It only has two states: HIGH and LOW, 1 and 0, +5 V and GND.

Step 4 — Audio Jack

Solder the **mini jack audio output**. The footprint on the PCB shows how to position it. The jack output, together with the speaker, forms the **speaking apparatus** of our little friend.





Step 5 — Arduino Headers (Brain Holders)

Add holders for the brain (Arduino) in the form of **pin headers**. This allows you to remove (transplant) the brain and use it for another project (patient).

If your Arduino comes without pre-soldered headers, solder them now by placing them into the **female headers** already attached to the PCB. Otherwise, skip this step.

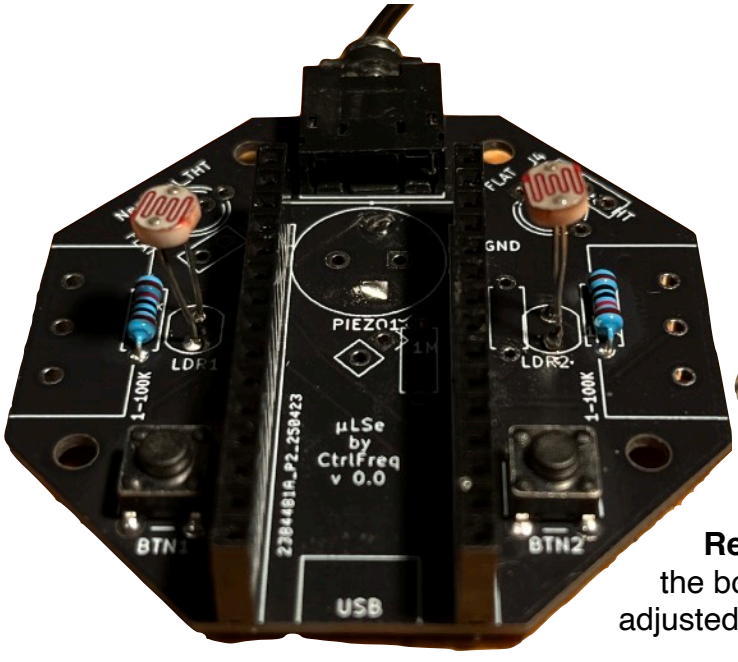
TIP for soldering headers (and other multi-pin components):

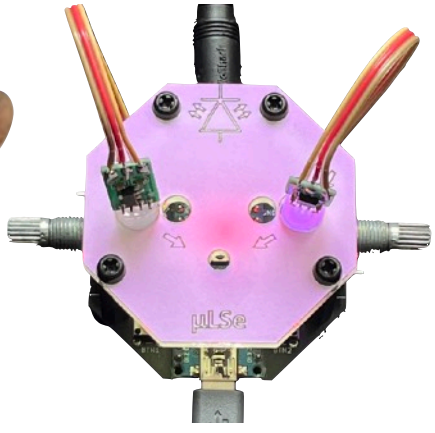
1. Solder a couple of pins first - this keeps the component in place and allows you to align it.
2. Then solder every other pin - this helps prevent excessive heat buildup.
3. Finally, solder the remaining pins.

Light Sensors = Eyes

Now comes a slightly tricky part — but fear not, there will be more!

It's quite sad to admit, but *μ*-LSe suffers from a bad case of *bulging eyes*. Its eyes (the light sensors) protrude about **10–15 mm** from their sockets — but, like aFrench bulldog, this adds a lot of charm.



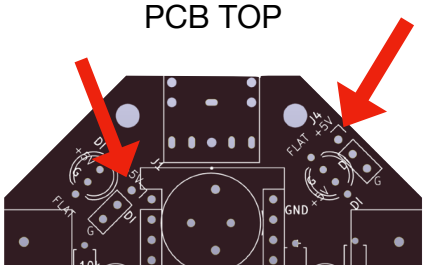


Cut and place the two **LDRs (Light Dependent Resistors)** so they stick out about **10–15 mm** from the board. It doesn't have to be perfect. Once adjusted, solder them in place (see reference picture).

Step 6 — RGB LEDs

Identify the two thin **3-core multi-cables** (about 10–15 cm long, colors may vary). Strip insulation from both ends, separate each core, and **tin** the wires with a tiny bit of solder — it'll make soldering and threading through holes easier.

PCB TOP



We're using LEDs with **3 pins: +5 V, DI, and GND**. Solder one side of each cable to the **3-pin points** on the PCB marked at the top left and right corners (see arrows). If there are no markings, the **square pad denotes +5 V**.

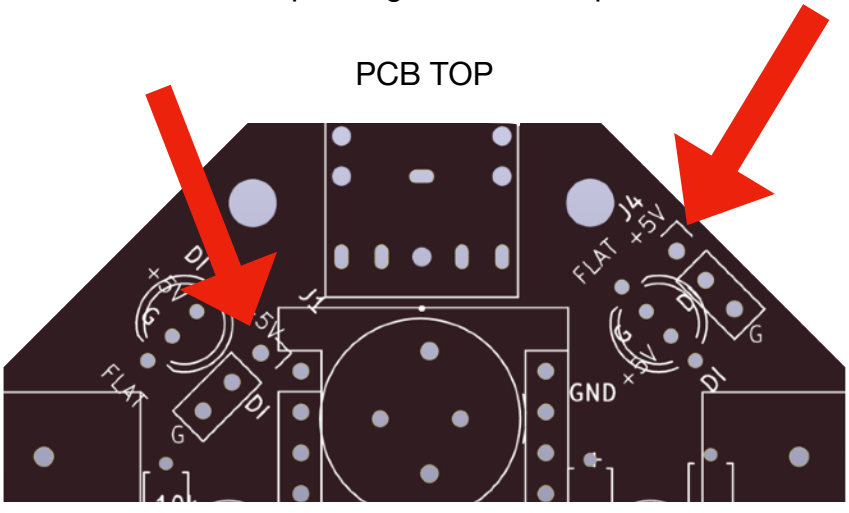
Now solder the other ends of the two cables to the **RGB LEDs**:

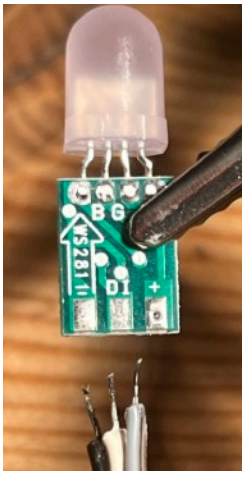
PCB SIDE	LED SIDE
+5 V (square pad)	+, 5V, 12V (depends on LED version)
DI (middle pad)	DI (make sure the side is correct)
GND (bottom pad)	G (GND, G, depends on LED version)

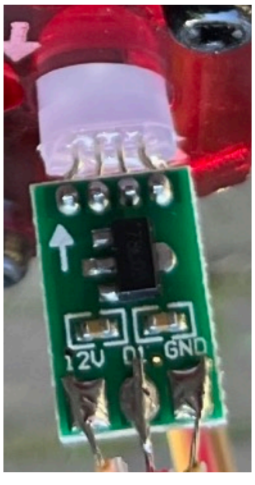
Make sure you solder to the **correct side of the LED board** - the one marked with **DI**. There are different LED versions, so double-check before soldering.

Arrows are pointing to the + / 5V pin

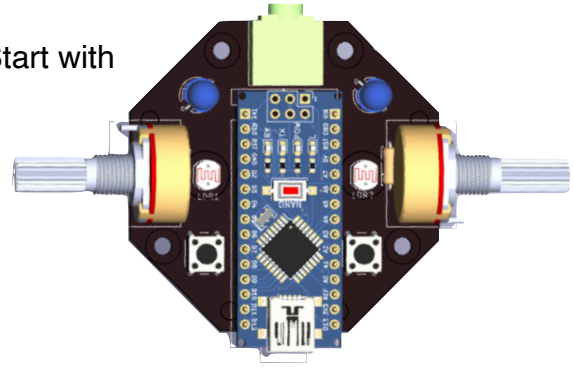
PCB TOP







Step 7 - Potentiometers



Solder the two **potentiometers** - one on each side. Start with the middle pin - it will help.

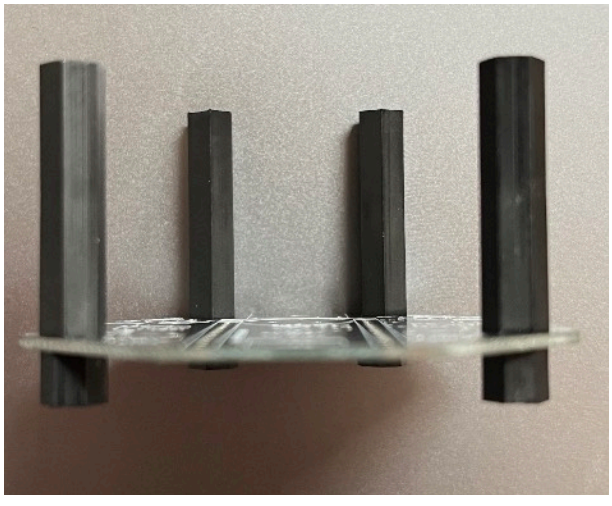
Light sensors and potentiometers are examples of **analog inputs** - unlike digital ones (buttons), they output continuous values between 0 V (GND) and 5 V.

Step 8 — Arduino

Plug the **Arduino** into the headers.

Congrats! You're almost there. If you're getting tired, take a break. :)

Step 9 - The Backbone



Identify the **4 short (5 mm)** and **4 long (25 mm)** hexagonal screws. Screw them together so that the short ones are at the bottom and the long ones on top — this **backbone** will hold all the parts together.

Whereas the PCB can be considered the skeleton, the Arduino is the **brain** of our little creature. In the next part, we'll learn how to reprogram (brainwash) it.

Step 10 - Skeleton & Body

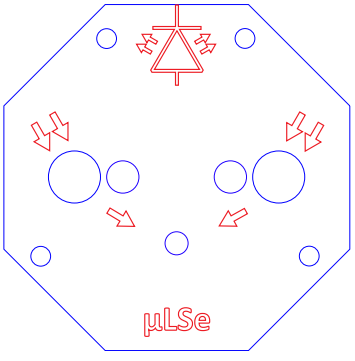
Identify the **top and bottom** (head and toes) of the skeleton. Make sure the **speaker cable** we soldered earlier goes through the hole in the bottom. Then solder the speaker from the **top**, as shown in the picture - otherwise, it won't fit.

You can also replace the speaker with a **piezo element** — it's a bit easier to attach to the bottom.

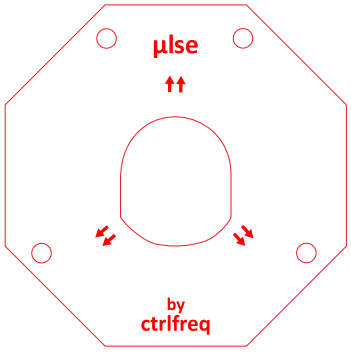
Use a **hot glue gun** or another adhesive to attach the speaker.

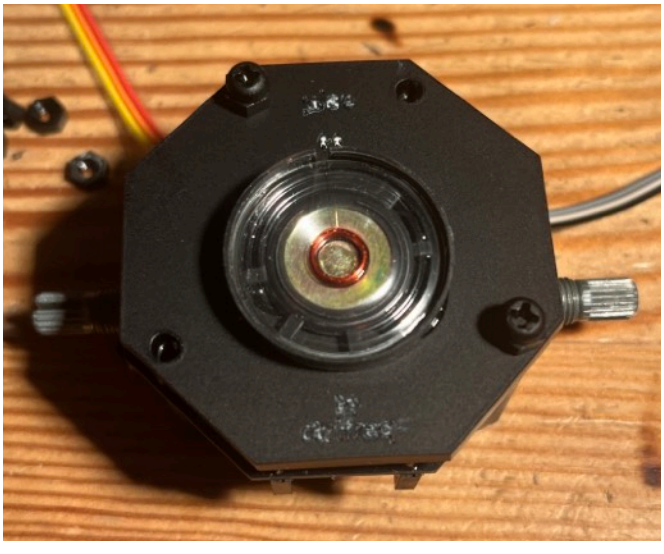
Be quick or be gone!

Top



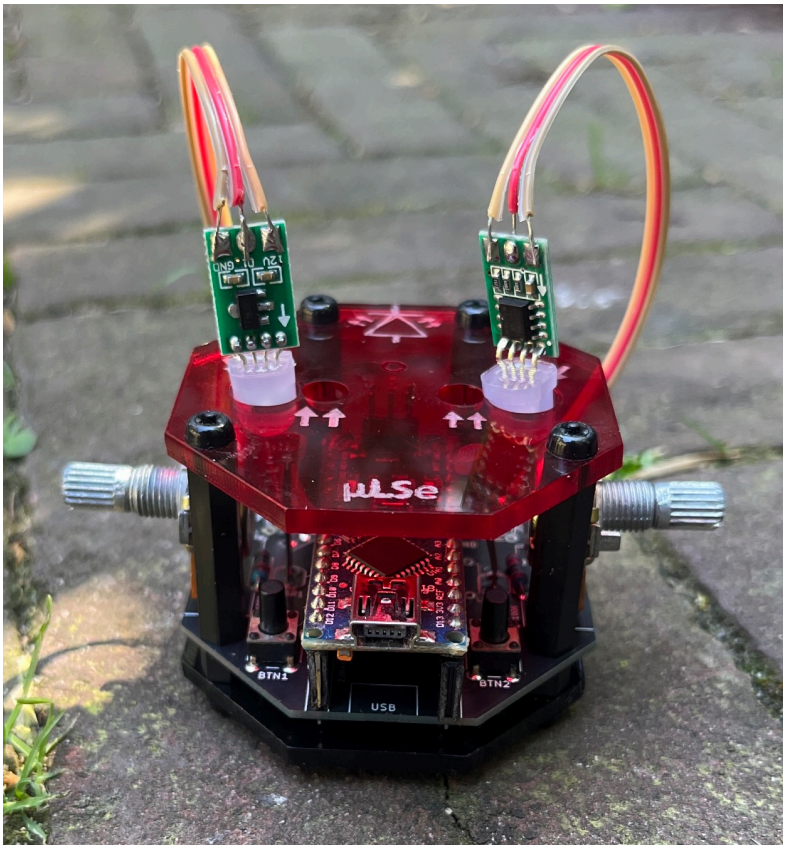
Bottom





Step 11 - Final Assembly

Attach the top and bottom of μ-LSe with the **8 small screws**. Your complete μ-LSe should now look more or less like this. If that's the case - Well done!!!



In the next part, we'll upload the code to the Arduino board. That's when your new companion will truly come to life.

CTRL FREQ

Losing Ctrl since 666!